Lab 8: Disk Scheduling

Suppose the head of a moving- head disk with 200 tracks, numbered 0 to 199 is currently serving request at tracks 143 and has finished a request at track 125. The queue it requests is kept in the FIFO order 86, 147, 91, 177, 94, 150, 102, 175, 130. Write a program to calculate the total head movement using following algorithms.

> - FCFS
> - SSTF
> - SCAN
> - C SCAN
> - LOOK
> - C LOOK

Codes and Outputs:

1. For FCFS

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int RQ[100], i, n, TotalHeadMovement = 0, initialHead;
    printf("Enter the number of Requests: ");
    scanf("%d", &n);
    printf("Enter the Requests sequence: ");
    for(i = 0; i < n; i++)
        scanf("%d", &RQ[i]);
    printf("Enter initial head position: ");
    scanf("%d", &initialHead);
    // FCFS disk scheduling
    for(i = 0; i < n; i++) {
        TotalHeadMovement += abs(RQ[i] - initialHead);
        initialHead = RQ[i];
    }
    printf("Total head movement is %d\n", TotalHeadMovement);
    return 0;
}
```

```
PS C:\Users\samir\Desktop\project> cd "c:\Users\samir\Desktop\project\" ; if ($?) { gcc te
e }
Enter the number of Requests: 9
Enter the Requests sequence: 86 147 91 177 94 150 102 175 130
Enter initial head position: 143
Total head movement is 565
PS C:\Users\samir\Desktop\project>
```

2. For SSTF

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
int RQ[100], i, n, TotalHeadMoment = 0, initial, count = 0;
printf("Enter the number of Requests: ");
scanf("%d", &n);
printf("Enter the Requests sequence: ");
for (i = 0; i < n; i++) {
scanf("%d", &RQ[i]);
}
printf("Enter initial head position: ");
scanf("%d", &initial);
// logic for sstf disk scheduling
/* loop will execute until all process is completed*/
while (count != n) {
int min = 1000, d, index;
for (i = 0; i < n; i++) {
d = abs(RQ[i] - initial);
if (min > d) {
min = d;
index = i;
}
}
TotalHeadMoment += min;
initial = RQ[index];
// 1000 is for max
// you can use any number
RQ[index] = 1000;
count++;
}
printf("Total head movement is %d\n", TotalHeadMoment);
return 0;
}
```

```
PS C:\Users\samir\Desktop\project> cd "c:\Users\samir\Desktop\project\" ; if ($?) { gcc tempCodeRu
e }
Enter the number of Requests: 9
Enter the Requests sequence: 86 147 91 177 94 150 102 175 130
Enter initial head position: 143
Total head movement is 162
PS C:\Users\samir\Desktop\project>
```

3. For SCAN

```
#include <stdio.h>
#include <stdlib.h>
int main() {
int RQ[100], i, j, n, TotalHeadMoment = 0, initial, size, move;
printf("Enter the number of Requests: ");
scanf("%d", &n);
printf("Enter the Requests sequence: ");
for (i = 0; i < n; i++) {
scanf("%d", &RQ[i]);
}
printf("Enter initial head position: ");
scanf("%d", &initial);
printf("Enter total disk size: ");
scanf("%d", &size);
printf("Enter the head movement direction for high 1 and for low 0: ");
scanf("%d", &move);
// logic for Scan disk scheduling
/* logic for sort the request array */
for (i = 0; i < n; i++) {
for (j = 0; j < n - i - 1; j++) {
if (RQ[j] > RQ[j+1]) {
int temp = RQ[j];
RQ[j] = RQ[j+1];
RQ[j+1] = temp;
}
}
}
int index;
for (i = 0; i < n; i++) {
if (initial < RQ[i]) {
index = i;
break;
}
}
// if movement is towards high value
if (move == 1) {
for (i = index; i < n; i++) {
TotalHeadMoment += abs(RQ[i] - initial);
```

```
initial = RQ[i];
}
// last movement for max size
TotalHeadMoment += abs(size - RQ[i-1] - 1);
initial = size - 1;
for (i = index - 1; i >= 0; i--) {
TotalHeadMoment += abs(RQ[i] - initial);
initial = RQ[i];
}
}
// if movement is towards low value
else {
for (i = index - 1; i >= 0; i--) {
TotalHeadMoment += abs(RQ[i] - initial);
initial = RQ[i];
}
// last movement for min size
TotalHeadMoment += abs(RQ[i+1] - 0);
initial = 0;
for (i = index; i < n; i++) {
TotalHeadMoment += abs(RQ[i] - initial);
initial = RQ[i];
}
}
printf("Total head movement is %d\n", TotalHeadMoment);
return 0;
}
```

```
PS C:\Users\samir\Desktop\project> cd "c:\Users\samir\Desktop\project\" ; if ($?) { gcc lab6.c
Enter the number of Requests: 9
Enter the Requests sequence: 86 147 91 177 94 150 102 130
130
Enter initial head position: 143
Enter total disk size: 200
Enter the head movement direction for high 1 and for low 0: 1
Total head movement is 169
PS C:\Users\samir\Desktop\project>
```

4. For C-SCAN

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int RQ[100], i, j, n, TotalHeadMovement = 0, initial, size, move;
    printf("Enter the number of Requests: ");
    scanf("%d", &n);
    printf("Enter the Requests sequence: ");
```

```
    for (i = 0; i < n; i++)
        scanf("%d", &RQ[i]);
    printf("Enter initial head position: ");
    scanf("%d", &initial);
    printf("Enter total disk size: ");
    scanf("%d", &size);
    printf("Enter the head movement direction for high 1 and for low 0: ");
    scanf("%d", &move);

    // Sort the request array
    for (i = 0; i < n; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (RQ[j] > RQ[j + 1]) {
                int temp = RQ[j];
                RQ[j] = RQ[j + 1];
                RQ[j + 1] = temp;
            }
        }
    }

    int index;
    index = n; // Default to n if initial is greater than all requests
    for (i = 0; i < n; i++) {
        if (initial < RQ[i]) {
            index = i;
            break;
        }
    }

    // If movement is towards high value
    if (move == 1) {
        for (i = index; i < n; i++) {
            TotalHeadMovement += abs(RQ[i] - initial);
            initial = RQ[i];
        }
        // Last movement for max size
        TotalHeadMovement += abs(size - RQ[i - 1] - 1);
        // Movement max to min disk
        TotalHeadMovement += abs(size - 1 - 0);
        initial = 0;
        for (i = 0; i < index; i++) {
            TotalHeadMovement += abs(RQ[i] - initial);
            initial = RQ[i];
        }
    }
    // If movement is towards low value
    else {
        for (i = index - 1; i >= 0; i--) {
```

```
            TotalHeadMovement += abs(RQ[i] - initial);
            initial = RQ[i];
        }
        // Last movement for min size
        TotalHeadMovement += abs(RQ[i + 1] - 0);
        // Movement min to max disk
        TotalHeadMovement += abs(size - 1 - 0);
        initial = size - 1;
        for (i = n - 1; i >= index; i--) {
            TotalHeadMovement += abs(RQ[i] - initial);
            initial = RQ[i];
        }
    }

    printf("Total head movement is %d\n", TotalHeadMovement);
    return 0;
}
```

```
PS C:\Users\samir\Desktop\project> cd "c:\Users\samir\Desktop\project\" ; if ($?) { gcc tempCode
e }
Enter the number of Requests: 9
Enter the Requests sequence: 86 93 46 99 126 56 145 133
126
Enter initial head position: 120
Enter total disk size: 200
Enter the head movement direction for high 1 and for low 0: 0
Total head movement is 392
PS C:\Users\samir\Desktop\project>
```

5. For LOOK

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int RQ[100], i, j, n, TotalHeadMoment = 0, initial, size, move;
    printf("Enter the number of Requests: ");
    scanf("%d", &n);
    printf("Enter the Requests sequence: ");
    for (i = 0; i < n; i++) scanf("%d", &RQ[i]);
    printf("Enter initial head position: ");
    scanf("%d", &initial);
    printf("Enter total disk size: ");
    scanf("%d", &size);
    printf("Enter the head movement direction for high 1 and for low 0: ");
    scanf("%d", &move);

    // Sort the request array
    for (i = 0; i < n - 1; i++) {
```

```
        for (j = 0; j < n - i - 1; j++) {
            if (RQ[j] > RQ[j + 1]) {
                int temp = RQ[j];
                RQ[j] = RQ[j + 1];
                RQ[j + 1] = temp;
            }
        }
    }

    int index = 0;
    for (i = 0; i < n; i++) {
        if (initial < RQ[i]) {
            index = i;
            break;
        }
    }

    // Movement towards high value
    if (move == 1) {
        for (i = index; i < n; i++) {
            TotalHeadMoment += abs(RQ[i] - initial);
            initial = RQ[i];
        }
        for (i = index - 1; i >= 0; i--) {
            TotalHeadMoment += abs(RQ[i] - initial);
            initial = RQ[i];
        }
    }
    // Movement towards low value
    else {
        for (i = index - 1; i >= 0; i--) {
            TotalHeadMoment += abs(RQ[i] - initial);
            initial = RQ[i];
        }
        for (i = index; i < n; i++) {
            TotalHeadMoment += abs(RQ[i] - initial);
            initial = RQ[i];
        }
    }

    printf("Total head movement is %d\n", TotalHeadMoment);
    return 0;
}
```

```
PS C:\Users\samir\Desktop\project> cd "c:\Users\samir\Desktop\project\" ; if ($?) { gcc tempCodeRunnerFi
e }
Enter the number of Requests: 9
Enter the Requests sequence: 22 97 64 53 78 129 25 167
189
Enter initial head position: 146
Enter total disk size: 200
Enter the head movement direction for high 1 and for low 0: 1
Total head movement is 210
PS C:\Users\samir\Desktop\project>
```

6. For C-LOOK

```
#include <stdio.h>
#include <stdlib.h>
int main() {
int RQ[100], i, j, n, TotalHeadMoment = 0, initial, size, move;
printf("Enter the number of Requests: ");
scanf("%d", &n);
printf("Enter the Requests sequence: ");
for (i = 0; i < n; i++) {
scanf("%d", &RQ[i]);
}
printf("Enter initial head position: ");
scanf("%d", &initial);
printf("Enter total disk size: ");
scanf("%d", &size);
printf("Enter the head movement direction for high 1 and for low 0: ");
scanf("%d", &move);
// logic for C-look disk scheduling
/*logic for sort the request array */
for (i = 0; i < n; i++) {
for (j = 0; j < n - i - 1; j++) {
if (RQ[j] > RQ[j + 1]) {
int temp;
temp = RQ[j];
RQ[j] = RQ[j + 1];
RQ[j + 1] = temp;
}
}
}
int index;
for (i = 0; i < n; i++) {
if (initial < RQ[i]) {
index = i;
break;
}
}
// if movement is towards high value
if (move == 1) {
for (i = index; i < n; i++) {
```

```
TotalHeadMoment = TotalHeadMoment + abs(RQ[i] - initial);
initial = RQ[i];
}
for (i = 0; i < index; i++) {
TotalHeadMoment = TotalHeadMoment + abs(RQ[i] - initial);
initial = RQ[i];
}
}
// if movement is towards low value
else {
for (i = index - 1; i >= 0; i--) {
TotalHeadMoment = TotalHeadMoment + abs(RQ[i] - initial);
initial = RQ[i];
}
for (i = n - 1; i >= index; i--) {
TotalHeadMoment = TotalHeadMoment + abs(RQ[i] - initial);
initial = RQ[i];
}
}
printf("Total head movement is %d\n", TotalHeadMoment);
return 0;
}
```

```
PS C:\Users\samir\Desktop\project> cd "c:\Users\samir\Desktop\project\" ; if ($
Enter the number of Requests: 9
Enter the Requests sequence: 22 97 64 53 78 129 25 167 189
Enter initial head position: 146
Enter total disk size: 200
Enter the head movement direction for high 1 and for low 0: 1
Total head movement is 317
PS C:\Users\samir\Desktop\project>
```